

Yunqian Zhang

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Satellite methane observations · atmospheric inverse modeling · machine learning for Earth observation

EDUCATION

Massachusetts Institute of Technology <i>PhD in Social and Engineering Systems, Institute for Data, Systems, and Society</i> Advisor: Prof. Daniel Varon	Cambridge, MA 2026-Present
Beijing Normal University <i>BSc in Environmental Science, School of Environment</i> GPA: 3.86/4.00	Beijing, China 2022-2026
University of California, Berkeley <i>Exchange Student, Berkeley Global Access Program</i> GPA: 4.00/4.00	Berkeley, CA 2024-2025

RESEARCH APPOINTMENTS

Massachusetts Institute of Technology, IDSS <i>PhD Researcher; advisor: Prof. Daniel Varon</i>	Cambridge, MA Aug 2026-Present
Beginning doctoral research on satellite methane monitoring and atmospheric inverse modeling, with emphasis on machine-learning-accelerated inversion and scalable Earth-observation workflows.	
University of California, Berkeley <i>Research Assistant; Prof. Lu Liang</i>	Berkeley, CA Aug 2024-Apr 2026
<i>PurpleAir temperature calibration.</i> Built a nationwide calibration pipeline using 797,744 hourly observations and tested temporal, spatial, and climate-zone transferability. - Developed XGBoost models that reduced mean absolute error to 0.64 °C in the best national temporal setting, yielding more than 88% error reduction; first-authored the resulting preprint. <i>Tree-crown delineation.</i> Integrated LiDAR-derived prompts with SAM 2 for zero-shot individual tree-crown segmentation and comparative evaluation.	
Harvard Kennedy School <i>Research Assistant; Prof. Charles Taylor</i>	Cambridge, MA Dec 2024-2026
- Constructed a county-year panel (2008-2024) linking USDA program participation, crop production, land cover, and remote-sensing indicators to study rice-conservation policy and methane-relevant outcomes. - Built geospatial and causal-analysis workflows for 39,045 county-year records, covering harmonization, treatment definition, matching diagnostics, and robustness checks.	
Massachusetts Institute of Technology, EAPS <i>Visiting Research Student; Prof. Edward A. Boyle</i>	Cambridge, MA Jun-Aug 2024; Jul-Sep 2025
- Supported marine geochemistry research through trace-metal clean-lab preparation, chromatographic separation, and MC-ICP-MS measurements. - Synthesized evidence on trace-metal cycling and organized laboratory data for reproducible analysis.	
Stony Brook University <i>Research Assistant; Prof. Minghao Qiu</i>	Remote May 2025-2026
- Integrated 22 years of CAMS reanalysis, MODIS aerosol products, land cover, and meteorology across 1,336 spatial grids to characterize dust exposure patterns. - Combined random-forest attribution with panel-regression designs to separate meteorological variability from land-use and socioeconomic correlates.	

PUBLICATIONS AND PREPRINTS

1. Zhang, Y., Rong, Y., & Liang, L. (2026). Nationwide evaluation and calibration of PurpleAir temperature sensors for urban thermal environment research. *SSRN Electronic Journal*. [Preprint] [doi](#)

2. Ma, M., Zhang, Y., Ma, Q., Wang, Z., Du, Z., Chen, Y., Gao, Q., Wang, F., Gao, B., & Sun, K. (2026). Fourteen-year field evidence reveals superior co-benefits of biochar in immobilizing heavy metals and sequestering carbon. *Biochar*, 8, 51. [doi](#)

3. Zhu, Y., Locke, W., Yuan, J., Zhang, Y., Ma, Q., & Liang, L. (2025). LiDAR-augmented zero-shot tree crown segmentation using SAM 2. *Information Geography*, 100025. [doi](#)

4. Chen, Y., Wang, Z., Zhang, A., Li, Y., Sun, K., Jiang, C., Gunina, A., Aloufi, A. S., Liang, X., Han, Z., Xiao, Y., Zhang, Y., Du, Z., Zhu, S., & Kuziyakov, Y. (2025). Polyethylene microplastics hamper aged biochar’s potential in mitigating greenhouse gas emissions. *Carbon Research*, 4, 5. [doi](#)

5. Liao, M., Gan, Z., Sun, W., Su, S., Li, Z., & Zhang, Y. (2024). Spatial distribution, source identification, and potential risks of 14 bisphenol analogues in soil from different functional zones in Chengdu, China. *Environmental Pollution*, 352, 124064. [doi](#)

ADDITIONAL RESEARCH

Beijing Normal University, Environmental Soil Chemistry

Undergraduate Researcher; Prof. Ke Sun

Beijing, China

Dec 2024-Jul 2026

- Analyzed 376 soil samples from a 14-year field experiment on biochar effects on heavy-metal immobilization and soil carbon sequestration.
- Applied structural equation modeling and variation partitioning; contributed data curation, validation, analysis, and writing to two peer-reviewed publications.

Beijing Normal University, Environmental Systems Analysis

Undergraduate Researcher; Prof. Xin Tian

Beijing, China

Aug 2023-Jul 2026

- Developed geospatial datasets and multi-criteria workflows for photovoltaic deployment and land-use optimization, integrating resource potential, environmental constraints, and infrastructure access.
- Co-authored a 47-page policy report on digital technologies and low-carbon development through literature synthesis, case-study coding, and cross-sector evidence review.

PRESENTATIONS

Human and Geographic Dimensions of Extreme Heat and Heat Risk. Poster presented at the NSF NCAR Research Symposium, Boulder, Colorado, June 2025.

HONORS AND AWARDS

2023, 2024	First Prize Scholarship, Beijing Normal University
2023, 2024	First Prize Undergraduate Incentive Scholarship, Beijing Normal University
2023	Second Prize, National Municipal Environmental Innovation Competition (team vice leader)
2023	Third Prize, National Undergraduate Physics Competition

METHODS AND TOOLS

Programming and modeling. Python (pandas, scikit-learn, XGBoost, CatBoost, SHAP), R, MATLAB

Geospatial and Earth observation. Google Earth Engine, ArcGIS Pro, QGIS; TROPOMI, GOSAT, ERA5, CAMS, MODIS

Laboratory. MC-ICP-MS, ICP-OES, ion chromatography, FTIR, SEM, BCR sequential extraction, DGT

Languages. Chinese (native); English (fluent)